

Implementation Strategy Intensity and Fidelity in School-Based Health Promotion: Results From a Cluster Randomized Effectiveness Trial

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ABSTRACT

Advancing widespread adoption of obesity prevention programs in schools requires understanding which implementation support strategies most effectively promote sustained program delivery. We randomized 45 school clusters to receive either standard implementation support or enhanced implementation support combining multiple reinforcement and linkage strategies over two academic years. Fidelity assessment revealed that approximately 75% of program components were delivered to the intended standards across both implementation arms. The enhanced support bundle did not produce significantly greater component delivery adherence compared to standard support. Two specific enhanced strategies (provider educational sessions and intersectoral coordination) were adopted by only a minority of participating schools due to reported staffing constraints and competing priorities. In contrast, parental participation and receipt of core health messages were more strongly influenced by school demographics and family characteristics than by implementation strategy intensity. These findings highlight that implementation success is complex and contextual, requiring adaptive refinement rather than static strategy application.

Trial Context

Underlying cluster randomized hybrid effectiveness-implementation trial comparing standard school-based health promotion with Healthy School Start program with implementation strategy optimization.